N 397

Seat No. | | | | | | | |

2024 VII 24 1100 – N 397– MATHEMATICS (71) GEOMETRY—PART II (E)

(REVISED COURSE)

Time : 2 Hours **(Pages 11)** **Max. Marks : 40**

Note :—

(*i*) *All* questions are compulsory.

(*ii*) Use of calculator is not allowed.

(*iii*) The numbers to the right of the questions indicate full marks.

(*iv*) In case of MCQs [Q. No. **1(A)**] only the first attempt will be evaluated and will be given credit.

(*v*) For every MCQ, the correct alternative (A), (B), (C) or (D) with subquestion number is to be written as an answer.

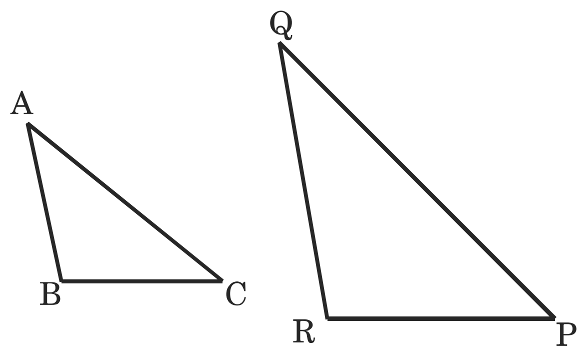
(*vi*) Draw proper figures of answers wherever necessary.

(*vii*) The marks of construction should be clear. Do not erase them.

(*viii*) Diagram is essential for writing the proof of the theorem.

1. **(A) For each of the following subquestions four alternative answers are given. Choose the correct alternative and write its alphabet:** **4**

(1) In ΔABC and ΔPQR, in a one to one correspondence of vertices, $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$, then .... statement is true.



(A) $\Delta PQR \sim \Delta ABC$

(B) $\Delta PQR \sim \Delta CAB$

(C) $\Delta CBA \sim \Delta PQR$

(D) $\Delta BCA \sim \Delta PQR$

(2) Two circles of radii 5.5 cm and 3.3 cm respectively touch each other externally. Then the distance between their centres is ....

(A) 4.4 cm

(B) 8.8 cm

(C) 2.2 cm

(D) 8.3 cm

(3) Seg AB is parallel to Y-axis and co-ordinates of point A are (1, 3), then the co-ordinates of point B are ....

(A) (3, 1)

(B) (5, 3)

(C) (3, 0)

(D) (1, –3)

(4) The volume of a cube of side 10 cm is ....

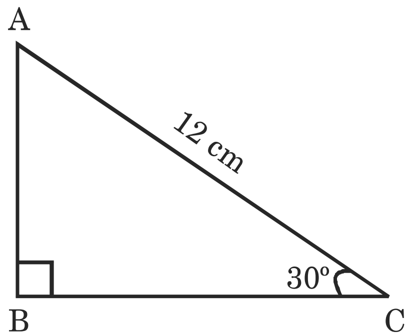
(A) 1000 cm^3

(B) 100 cm^3

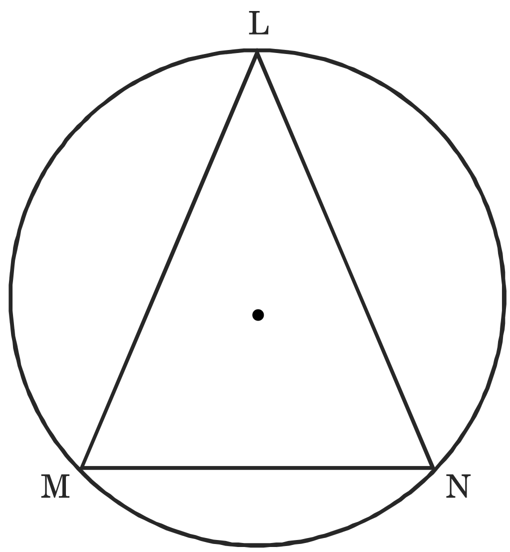
(C) 10,000 cm^3

(D) 10 cm^3

(B) **Solve the following subquestions :** **4**

(1) 

In ΔABC, $\angle B = 90^\circ$, $\angle C = 30^\circ$, $AC = 12$ cm, then find AB.

(2) 

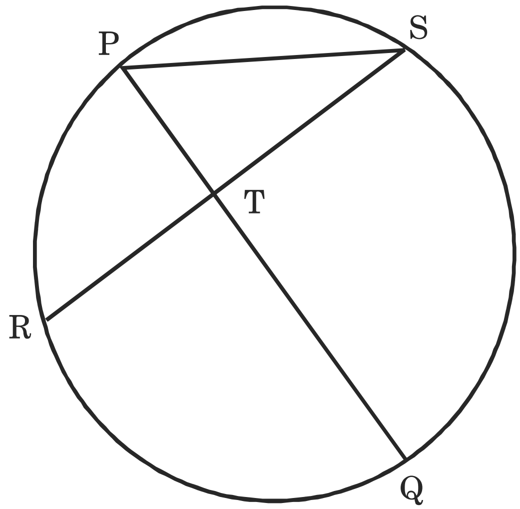
In the above figure, $m(\text{arc } MN) = 70^\circ$, find $\angle MLN$.

(3) Find the value of $\sin \theta \times \operatorname{cosec} \theta$.

(4) If radius of a circle is 4 cm and length of an arc is 10 cm, then find the area of the sector.

2. (A) Complete the following activities and rewrite it (any *two*) : 4

(1)



In the above figure, chord PQ and chord RS intersect at point T. Then complete the following activity to prove,

$\angle STQ = \frac{1}{2}[m(\text{arc SQ}) + m(\text{arc PR})]$

Activity :

$\angle STQ = \angle SPQ + \boxed{}$ .... (Remote interior angles theorem of a triangle)

$= \frac{1}{2} m(\text{arc SQ}) + \boxed{}$ .... (Inscribed angle theorem)

$= \frac{1}{2}\left[\boxed{} + \boxed{}\right]$

P.T.O.

(2) If $\sec \theta = \frac{25}{7}$, complete the following activity to find the value of $\tan \theta$.

Activity :

We have,

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\therefore \quad 1 + \tan^2 \theta = \boxed{}$$

$$\therefore \quad \tan^2 \theta = \frac{625}{49} - \boxed{}$$

$$\therefore \quad \tan^2 \theta = \frac{625 - 49}{49}$$

$$\therefore \quad \tan^2 \theta = \frac{\boxed{}}{49}$$

$$\therefore \quad \tan \theta = \boxed{}$$

(3) Complete the following activity to find the volume of a cone if the radius of its base is 7 cm and its perpendicular height is 6 cm.

Activity :

Here, radius (r) = 7 cm

height (h) = 6 cm

$\therefore$ Volume of cone = $\boxed{}$ .... (Formula)

$$= \frac{1}{3} \times \frac{22}{7} \times \boxed{} \times 6$$

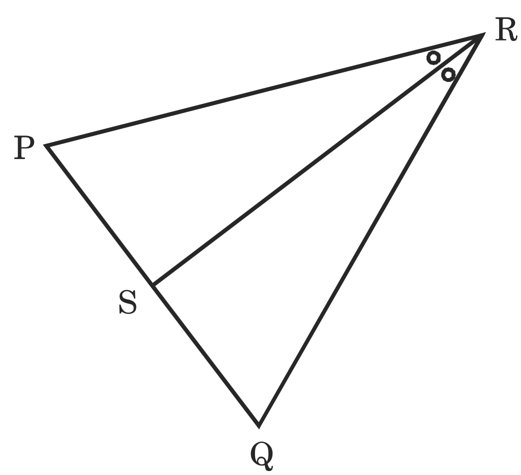
$$= \frac{1}{3} \times \frac{22}{7} \times \boxed{} \times 6$$

$\therefore$ Volume of cone = $\boxed{}$ cm^3

(B) Solve the following subquestions (any *four*) : 8

(1) Construct a tangent to a circle with centre P and radius 3.2 cm at any point M on it.

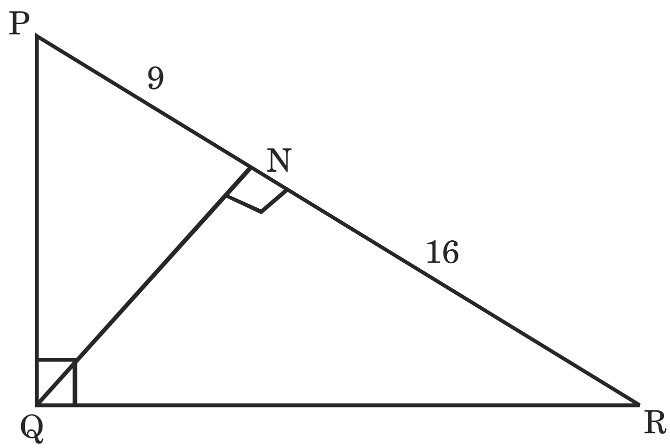
(2)



In ΔPQR, seg RS bisects $\angle PRQ$. If PR = 15, RQ = 20, PS = 12, then find SQ.

(3) Find the surface area of a sphere of diameter 14 cm.

(4)

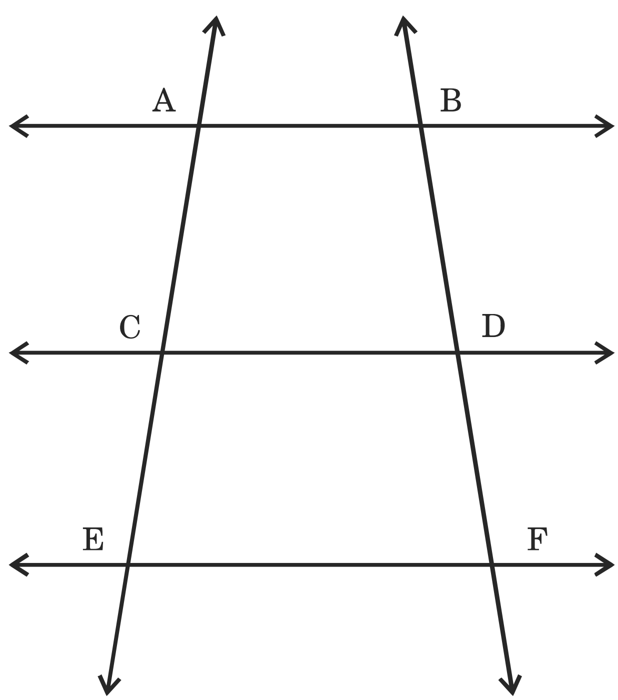


In the above figure, $\angle PQR = 90^\circ$, seg QN $\perp$ seg PR, PN = 9, NR = 16, find QN.

(5) Find the slope of the line passing through the points A(3, 3) and B(5, 7).

3. (A) Complete the following activities and rewrite it (any *one*) : 3

(1)



In the given figure, AB || CD || EF. If AC = 12, CE = 9, BD = 8, then complete the following activity to find DF.

Activity : AB || CD || EF

$\therefore \quad \dfrac{AC}{\boxed{}} = \dfrac{\boxed{}}{DF}$ .... (Property of three parallel lines and their transversal)

$\therefore \quad \dfrac{12}{\boxed{}} = \dfrac{\boxed{}}{DF}$

$\therefore \quad DF = \dfrac{8 \times 9}{\boxed{}}$

$\therefore \quad DF = \boxed{}$

(2) A cylinder is having radius 7 cm and height 21 cm. Complete the following activity to calculate its volume and circumference of its base.

Activity :

Radius (r) = 7 cm

Height (h) = 21 cm

Volume of cylinder = $\boxed{}$ .... (Formula)

$= \dfrac{22}{7} \times 7 \times 7 \times \boxed{}$

$= 154 \times 21$

$$= \boxed{} \text{ cm}^3$$

Circumference of base = $\boxed{}$ .... (Formula)

$$= 2 \times \frac{22}{7} \times \boxed{}$$

$$= \boxed{} \text{ cm}$$

(B) Solve the following subquestions (any *two*) : **6**

(1) Prove that, 'Opposite angles of a cyclic quadrilateral are supplementary'.

(2) Draw a circle with centre P and radius 3.5 cm. Take point Q at a distance 8 cm from the centre. Construct tangents to the circle from point Q.

(3) Show that points P(–2, 3), Q(1, 2), R(4, 1) are collinear.

(4) If $\Delta PQR \sim \Delta LMN$, $9 \times A(\Delta PQR) = 16 \times A(\Delta LMN)$ and QR = 20, then find MN.

4. Solve the following subquestions (any *two*) : **8**

(1) ΔABC is an equilateral triangle. Point D is on seg BC such that $BD = \frac{1}{5}$ BC. Then prove that $\frac{AD^2}{AB^2} = \frac{21}{25}$.

(2) $\Delta LMN \sim \Delta LQP$. In ΔLMN, LM = 3.6 cm, $\angle L = 50^\circ$, LN = 4.2 cm and $\frac{LM}{LQ} = \frac{4}{7}$. Construct ΔLQP.

(3) To find the width of the river, a man observes the top of a tree, on the opposite bank making an angle of elevation of 60º. When he moves 24 meter backward from bank and observes the same top of the tree, his line of vision makes an angle of elevation of 30º. Find the height of the tree and width of the river. ($\sqrt{3} = 1.73$)

5. Solve the following subquestions (any *one*) : 3

(1) The diameter of circular garden is 13 meter. The distance between two gates of garden is 13 meter. An electric pole is to be erected on the circumference of the garden such that the difference between distance of the pole from each gate is 7 meter. Can such pole be erected ? If yes, find the distance of the pole from each gate.

(2) The line $x - 6y + 11 = 0$ bisects the segment joining the points (8, –1) and (0, k), then find the value of k.